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ACADEMY OF TECHNOLOGY
INTERNAL EXAMINATION (CA1), ODD SEM, 2026

Programme/Discipline: CSE
Paper Name: Compiler Design
Date of Examination: 27/08/2026

Paper Code: PCC-CS501
Duration: 1 Hour

Semester: 5th
UPID: 5506
Full Marks: 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO1	CO1	CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO1	CO1	CO2	CO2
Bloom's Level	L1	L1	L1	L1	L1	L1	L3	L3	L2	L3	L1	L3	L1

Group A: Attempt any five questions from the following **[1x5=5]**

1. i. Compiler has _____ phases?
   a) 4 b) 5 c) 6 d) 7

   ii. Which of the following is known as a compiler for a high-level language that runs on one machine and produces code for a different machine?
   a) Cross compiler b) Multipass compiler c) Optimizing compiler d) One-pass compiler

   iii. What is the linker?
   a) It is always used before the program execution b) It is required to create the load module
   c) It is the same as the loader d) None of the above

   iv. Which of the following is a stage of compiler design?.
   a) Semantic analysis b) Intermediate code generator c) Code optimizer d) All of the mentioned .

   v. Which of the following groups is/are token together into semantic structure ? .
   a) Syntax Analyzer b) Intermediate code generation
   c) Lexical Analyzer d) Semantic Analyzer

   vi. Compiler should report the presence of _____ in the source program in translation process.
   a) classes b) objects
   c) errors d) text .

   vii. Count the number of tokens in the following: printf("Value of x=%d", x);
   a) 12 b) 11 c) 13 d) 7

Group B: Attempt any four questions from the following. **[4x5=20]**

2. Convert the regular expression (a|b)*abb to DFA directly . [5]
3. a) Count the number of tokens in the following: main() { a= b+++---+++===; printf("%d%d",a,b);}
   b) What are Multi-pass Compiler and One-pass Compiler? [3+2]
4. Generate target code for the following :
   sum= old_sum + rate*50;
   Show every stages. [5]
5. a) Write down the functions of Linker and Loader
   b) Define Compiler [4+1]
6. Construct an NFA (Using Thomson's Algorithm) for the following Regular Expression (a|b)*abb (a|b)*
   Convert the NFA to equivalent DFA and minimize the number of steps of the DFA. [5]
7. Clarify the following terms with suitable examples (i) Token (ii) Lexeme (iii) Pattern [5]

ACADEMY OF TECHNOLOGY
Internal Examination (CA1), Odd Semester, 2026

Programme/Discipline: B.Tech/ CSE **Semester:5th**
Paper Name: Operating System **Paper Code: PCC-CS502** **UPID: 5507**
Date of Examination: 27/08/2026 **Duration: 1 Hour** **Full Marks: 25**

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1. vii	2	3	4	5	6	7
CO Mapping	CO1	CO1	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2
Bloom's Level	L1	L1	L1	L1	L2	L1	L1	L3	L2	L2	L2	L2	L2

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i. In system call implementation, the system-call interface uses a table indexed by:
a) Process IDs b) A number associated with each system call
c) File descriptors d) Memory addresses

ii. Which of the following is an OS service primarily helpful to the user, rather than for efficient system operation?
a) Resource allocation b) Accounting c) Program execution d) Protection and security

iii. A process is best defined as:
a) A program stored on disk b) A program in execution
c) A collection of PCBs d) A scheduling queue

iv. When a process is waiting to be assigned to a processor, it is in the _____ state.
a) New b) Running c) Waiting d) Ready

v. Which of the following is an advantage of cooperating processes?
a) Increased context-switch overhead b) Computation speed-up
c) Reduced modularity d) Elimination of the need for IPC

vi. The scheduler that selects which process should be executed next and allocates the CPU is the:
a) Long-term scheduler b) Short-term scheduler
c) Medium-term scheduler d) I/O scheduler

vii. The preemptive version of Shortest-Job-First scheduling is also known as:
a) Shortest-Remaining-Time-First (SRTF) b) Round Robin
c) Multilevel Feedback Queue d) FCFS

Group B

Attempt any four questions from the following. [5 × 4 = 20]

2. Consider the 3 processes, P1, P2 and P3are shown in the table
   <Process, Arrival time, Burst Time> = {P1, 0, 5}{P2,1,7}{P3, 3, 4}
   Draw the Gantt chart and calculate the average waiting time and average turnaround time using Preemptive SJF (Shortest Job First) and Nonpreemptive SJF algorithms. 3+2
3. Define the five scheduling criteria (CPU utilization, throughput, turnaround time, waiting time, response time) and state the optimization goal (max/min) for each. Explain why response time is emphasized in interactive systems while turnaround time matters more for batch systems.
4. Define a process and explain the different states a process passes through during its lifetime, with the help of a state diagram.
5. What is a Process Control Block (PCB)? List and explain the information it stores, and describe its role during a context switch.
6. What is context switching? Why is it considered to be an overhead?

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ACADEMY OF TECHNOLOGY
Internal Examination (CA1), Odd Sem, 2026

Discipline: CSE **Sem: 5** **Paper Name:** Object Oriented Programming **Paper Code:** PCC CS503
UPID: 5504 **Date of Examination:** 28/08/2026 **Duration:** 1 Hour **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2. a, b	3	4. a, b	5	6. a, b	7. a, b
CO Mapping	CO1	CO1	CO1	CO1	CO2	CO1	CO1	CO1, CO1	CO2	CO1, CO1	CO1	CO1, CO1	CO1, CO1
Bloom's Level	L2	L2	L1	L2	L3	L1	L2	L2, L2	L3	L2, L2	L2	L2, L2	L2, L2

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

(i) Which of the following primarily describes an Abstract Data Type (ADT)?

a. The internal memory representation of data
b. The implementation language used to develop a data structure
c. The behaviour and operations of a data type independent of its implementation
d. The physical memory location of an object

(ii) Which OOP concept combines data and the methods that operate on that data into a single unit?

a. Inheritance b. Polymorphism c. Encapsulation d. Abstraction

(iii) Which keyword is used to refer to the current object inside an instance method or constructor?

a. super b. self c. this d. current

(iv) What is the purpose of this() inside a constructor?

a. It creates a new object b. It calls another constructor of the same class
c. It calls a method of another class d. It returns the current object

(v) A Department object contains several Student objects, but a student can exist independently even if the department is removed. Which relationship is represented?

a. Composition b. Aggregation c. Inheritance d. Uses-a

(vi) Which access modifier provides the highest level of access restriction for a class member?

a. public b. protected c. Default d. private

(vii) Which of the following is an example of method overloading?

a. void add(int a) and void add(int a, int b) b. void add(int a) and int add(int a)
c. void add(int a) and void subtract(int a) d. int add(int a) and void add(int a)

Group B

Attempt any four questions from the following. [5 x 4=20]

2. a) Explain the relationship between a class, object, and method. b) What is the difference between passing a primitive value and passing an object through method? Explain. [3 + 2]

3. Create a class Distance with feet and inches. Write methods to input, add two distance objects and display the result. [5]

4. a) What is the difference between a constructor and a method? b) Can constructors be overloaded? Explain [3 + 2]

5. Explain the significance of each token of "*public static void main (String args[])*". [5]

6. a) What is the difference between static and instance variables? b) What happens to a static variable when multiple objects are created? [3 + 2]

7. a) Explain constructor chaining with an example. (b) Explain how a method can return an object? [3 + 2]

ACADEMY OF TECHNOLOGY
INTERNAL EXAMINATION (CA1), ODD SEMESTER, 2026

Program/Discipline: CSE
Paper Name: Software Engineering
Date of Examination: 28/08/2026
Paper Code: ESC501
Duration: 1 Hour
Semester: 5th
UPID: 5505
Full Marks: 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO1	CO2	CO1	CO4	CO2	CO4	CO2	CO1	CO2	CO4	CO4	CO2	CO4
Bloom's Level	L1	L2	L1	L4	L2	L4	L2	L1	L2	L4	L4	L2	L4

Group A

Attempt any five questions from the following:

[1 × 5 = 5]

i) Define Software Engineering?

ii) SDLC stands for _____.

iii) What are the different phases in SDLC?

iv) Write two important differences between program and Software.

v) In which model, a 'Toy Model' is developed before the final development of the product.

vi) Write the main difference between V- Model and Waterfall model.

vii) Different types of Risks are properly managed in _____ model.

Group B

Attempt any four questions from the following:

2. Define SDLC. Describe briefly various stages of SDLC. [5]

3. Discuss important properties of the Spiral model. Why is the Spiral model called the Meta model? [5]

4. A client is unable to clearly specify the complete requirements of a system. Analyze which SDLC model you would recommend? [5]

5. Compare and contrast the Prototype model and the Evolutionary model. [5]

6. Explain the RAD (Rapid Application Development) Model and discuss its major advantages and limitations. [5]

7. Analyze how the Iterative Model helps developers accommodate changing requirements during software development. [5]

ACADEMY OF TECHNOLOGY

Internal Examination (CA1) Odd Sem, 2026

Programme/Discipline: B.TECH , CSE Semester: 5th

Paper Name: Introduction to Industrial Management Paper Code: HSMC 501 UPID: 5508

Date of Examination: 29/08/26 Duration: 1 Hour Full Marks:25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1.vii	2	3	4	5	6	7
CO Mapping	1	1	1	1	1	1	1	1	1	1	1	1	1
Bloom's Level	1	1	1	2	1	2	1	1	2	1	2	1	2

Group A

1 Attempt any five questions from the following [1 × 5 = 5]

i. A system is best defined as

a) A random collection of objects b) A single independent unit

c) A group of interrelated components working together to achieve a goal d) A set of unrelated activities

ii. Which of the following is not a function of management?

a) Planning b) Organizing c) Guiding d) controlling

iii. The primary objective of an organizational structure is to

a) Increase paperwork b) Define roles, responsibilities and authority

c) Fix employee salaries d) Eliminate communication

iv. Which organizational structure is most suitable for a manufacturing company producing a single product

a) Matrix Structure b) Functional Structure

b) Divisional structure d) Project Structure

v. Delegation of authority refers to:

a) Transferring authority and responsibility to subordinates while retaining accountability

b) Giving up all managerial responsibilities permanently

c) Eliminating the need for supervision

d) Concentrating all authority at the top level

vi. A company allows departmental managers to make routine decisions without approval from top management. This practice is known as

a) Centralization b) Decentralization c) Scalar process d) Division of labor

vii. Which statement best describes organizational culture?

a) It refers to the physical layout of the workplace.

b) It consists of the shared values, beliefs, and norms of an organization.

c) It measures employee attendance.

d) It represents the annual financial performance of the organization.

Group B

Attempt any four questions from the following.

2. Write down the functions of management. [5]
3. Write a short note on span of control. [5]
4. Explain the major factors influencing job satisfaction in an organization. [5]
5. a) What do you mean by organization structure? [2]
   b) What are the types of organization structures? [3]
6. Discuss the factors affecting organizational culture. [5]
7. Explain the differences between centralization and decentralization in industrial management with suitable examples. [5]

AOT

ACADEMY OF TECHNOLOGY
Internal Examination (CA1/CA2), Odd/Even Sem, 2026

Programme/Discipline: Computer Science and Engineering | Semester: 5th

Paper Name: Artificial Intelligence | Paper Code: PEC-IT501B | UPID: 005510

Date of Examination: 29/08/2026 | Duration: 1 Hour | Full Marks: 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1. vii	2	3	4	5	6	7
CO Mapping	1	1	1	1	1	1	1	1	1	1	1	1	1
Bloom's Level	1	1	1	1	1	1	1	2	2	2	2	3	2

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i. Which element in the agent is used for selecting external actions?
a) Perceive b) Performance c) Learning d) Actuator

ii. Which agent deals with happy and unhappy states?
a) Simplex reflex agent b) Model based agent c) Learning agent d) Utility based agent

iii. The action of the Simple reflex agent completely depends upon _____
a) Perception history b) Current perception c) Learning theory d) Utility functions

iv. What is the expansion of PEAS in the task environment?
a) Peer, Environment, Actuators, Sense b) Perceiving, Environment, Actuators, Sensors
c) Performance, Environment, Actuators, Sensors d) None of the mentioned

v. Which is used to provide feedback to the learning element?
a) Critic b) Actuators c) Sensor d) None of the mentioned

vi. Which of the following is not a type of Artificial Intelligence agent?
a) Learning AI agent b) Goal-based AI agent
c) Simple reflex AI agent d) Unity-based AI agent

vii. What kind of environment is a crossword puzzle?
a) Static b) Dynamic c) Semi Dynamic d) None of the mentioned

Group B

Attempt any four questions from the following. [5 × 4 = 20]

2. Discuss the environment for agents. Explain four types of environments related to intelligent agents. [1+4]
3. Explain the PEAS Representation Model with an example. [5]
4. Explain how AI can be used in fraud detection. [5]
5. Describe Goal-based Agents with a diagram. [5]
6. A farmer wants to cross a river and take with him a wolf, a goat and a cabbage. There is only one boat available which can support the farmer and either the goat, wolf or the cabbage. But the problem is, if the goat and wolf are left alone (either in the boat or onshore), the wolf will eat the goat. Similarly, if the Goat and cabbage are left alone, then the goat will eat the cabbage.
How can the farmer bring the wolf, the goat, and the cabbage across the river? Show the possible solution with the help of state space representation. [5]
7. Describe Learning Agents with a diagram. [5]